A 5,454-VERTEX MULTI-LABEL COUNTEREXAMPLE TO SEYMOUR–SULLIVAN CONJECTURE 6.1 AND PROTOTYPE OPTIMALITY

ARGUS OPEN RESEARCH PROJECT

Abstract. Let $G = (V, E)$ be a finite simple directed graph in which each directed arc carries a subset of labels from $\{1, \dots, k\}$, equivalent to a collection of simple directed graphs $G_1, \dots, G_k$ on the common vertex set V. Write $D(v) = \sum_{i=1}^{k} d^+_{G_i}(v)$ for the total outgoing label degree of v, and let $R(v)$ denote the set of vertices reachable from v by a nonempty rainbow directed walk under unordered injective label sequences. Conjecture 6.1 in Sullivan's survey asks whether every rainbow-cycle-free edge-labelled digraph contains a vertex v with $|R(v)| \geq D(v)$. Under the multi-label formulation where a single directed arc may belong to several label classes G_i and reachability is taken over positive-length rainbow paths, we construct an explicit 16-vertex prototype with $k = 6$ labels whose independent-set blow-up yields a directed graph on 5,454 vertices and 11,792,572 arcs containing no rainbow directed cycle and satisfying $D(v) - |R(v)| \geq 1$ at every vertex v. Using an explicit nonnegative integer Farkas dual pair, we further prove that 5,454 is the exact linear-programming and integer-programming minimum vertex count among all positive independent-set blow-ups of this 16-vertex prototype, with zero integrality gap. Finally, we prove that no counterexample exists with $n \leq 4$ vertices and $k \leq 3$ labels, nor in any positive independent-set blow-up of a template with $n \leq 4$ and $k \leq 3$.

1. Introduction and Problem Formulation

Let V be a finite vertex set and let $G_1 = (V, E_1), \dots, G_k = (V, E_k)$ be simple loopless directed graphs on V. Equivalently, each ordered pair (u, v) of distinct vertices carries a palette $L(u, v) = \{i \in \{1, \dots, k\} : (u, v) \in E_i\} \subseteq \{1, \dots, k\}$. For each vertex $v \in V$, the total outgoing label incidence is

$$D(v) \;=\; \sum_{i=1}^{k} d^+_{G_i}(v) \;=\; \sum_{w \in V \setminus \{v\}} |L(v, w)|.$$

A *rainbow directed walk* of length $\ell \geq 1$ from u_0 to u_ℓ is a sequence of arcs $(u_0, u_1), \dots, (u_{\ell-1}, u_\ell)$ together with a choice of pairwise distinct labels $c_r \in L(u_{r-1}, u_r)$ for $1 \leq r \leq \ell$. A *rainbow directed cycle* is a simple directed cycle that admits a rainbow labelling; since any closed rainbow walk contains a simple rainbow directed cycle, absence of rainbow directed cycles is equivalent to the absence of nonempty closed rainbow walks.

For $v \in V$, let $R(v)$ denote the set of endpoints $w \in V$ of nonempty rainbow directed walks starting at v. When G has no rainbow directed cycle, $v \notin R(v)$ under this positive-length reachability convention. In the survey on the Caccetta–Häggkvist conjecture and related problems, Sullivan [1] states Conjecture 6.1 (attributed to Seymour and Sullivan): either G contains a rainbow directed cycle, or there exists a vertex $v \in V$ with $|R(v)| \geq D(v)$. Equivalently, a counterexample requires a rainbow-cycle-free multi-label digraph in which every vertex $v \in V$ has strict deficit

$$\delta(v) \;:=\; D(v) - |R(v)| \;\geq\; 1.$$

Theorem 1.1 (Main Counterexample and Prototype Optimality)**.** *Under the positive-length, unordered injective-label, multi-label formulation above with $k = 6$ labels:*

(1) *There exists an explicit 16-vertex template T and positive integer vertex multiplicities $m = (m_0, \dots, m_{15}) \in \mathbb{Z}^{16}_{\geq 1}$ with*

$$m \;=\; (1096,\ 8,\ 23,\ 69,\ 244,\ 755,\ 1,\ 1059,\ 1,\ 1,\ 1131,\ 1,\ 1,\ 1,\ 1,\ 1062)$$

whose independent-set blow-up $G = T[m]$ has $\sum_{u=0}^{15} m_u = 5{,}454$ vertices, 11,792,572 directed arcs, and 24,094,765 labelled arc incidences, contains no rainbow directed cycle, and satisfies $D(v) - |R(v)| \geq 1$ for every $v \in V(G)$, with

$$\min_{v \in V(G)} \big(D(v) - |R(v)|\big) \;=\; 1.$$

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(2) *Among all positive real or integer multiplicity vectors $m \in \mathbb{R}^{16}_{\geq 1}$ on the template T satisfying $\delta_T(u; m) \geq 1$ for all $0 \leq u \leq 15$, the minimum total vertex count is exactly* 5,454*:*

$$\min \left\{ \sum_{u=0}^{15} m_u \;\middle|\; m \in \mathbb{R}^{16}_{\geq 1},\ Am \geq \mathbf{1} \right\} \;=\; 5{,}454,$$

and the integrality gap is zero.

Theorem 1.2 (Small-Template and Blow-Up Exclusions)**.** *For $k \leq 3$ labels and templates on $n \leq 4$ vertices:*

(1) *Exhaustive enumeration of all* 34,012,224 *digon-free 3-label patterns on $n \leq 4$ vertices leaves* 27,702 *rainbow-cycle-free patterns with at least one multiply labelled outgoing arc per vertex; every one has $\min_v(D(v) - |R(v)|) \leq 0$.*

(2) *Across all* 27,702 *retained templates on $n \leq 4$ vertices (which yield* 1,854 *distinct deficit matrices A), every linear system $Am \geq \mathbf{1},\ m \geq \mathbf{1}$ admits an explicit nonzero nonnegative integer dual vector $y \in \mathbb{Z}^n_{\geq 0} \setminus \{0\}$ with $A^{\mathsf{T}} y \leq 0$, proving that no positive independent-set blow-up of any $n \leq 4,\ k \leq 3$ template can satisfy $D(v) - |R(v)| \geq 1$ everywhere.*

2. Independent-Set Blow-Ups and Exact Deficit Linearization

Let T be a template on vertex set $\{0, 1, \ldots, n-1\}$ with label palettes $L_T(u, v) \subseteq \{1, \ldots, k\}$ and capacity matrix $C_{u,v} = |L_T(u, v)|$. Given positive integers $m = (m_0, \ldots, m_{n-1}) \in \mathbb{Z}^n_{\geq 1}$, the *independent-set blow-up* $G = T[m]$ replaces each template vertex u by an independent set V_u of m_u clone vertices, and places between every $x \in V_u$ and $y \in V_v$ an arc carrying the exact palette $L_T(u, v)$.

Lemma 2.1 (Blow-Up Projection and Linear Deficit Formula)**.** *Let T be a template and $G = T[m]$ its blow-up for $m \in \mathbb{Z}^n_{\geq 1}$.*

(a) *Every rainbow walk in G projects under $V_u \mapsto u$ to a rainbow walk in T with the same label sequence, and every rainbow walk in T lifts to every choice of clones along its vertices. In particular, G is rainbow-cycle-free if and only if T is rainbow-cycle-free.*

(b) *Assume T is rainbow-cycle-free, and let $R_T(u) \subseteq \{0, \ldots, n-1\} \setminus \{u\}$ denote the set of template vertices reachable from u by a nonempty rainbow walk in T. Then for every clone $x \in V_u$,*

$$D_G(x) \;=\; \sum_{v=0}^{n-1} C_{u,v}\, m_v, \qquad |R_G(x)| \;=\; \sum_{v \in R_T(u)} m_v,$$

and consequently $D_G(x) - |R_G(x)| = (Am)_u$, where the $n \times n$ integer matrix A is given by

$$A_{u,v} \;=\; C_{u,v} - \mathbf{1}_{\{v \in R_T(u)\}} \qquad (0 \leq u, v < n).$$

Proof. Since each V_u is an independent set and every pair $(x, y) \in V_u \times V_v$ carries the identical palette $L_T(u, v)$, the map sending $x \in V_u$ to u is a label-preserving homomorphism whose fibers are independent sets with complete bipartite inter-fiber connections. Thus rainbow walks correspond bijectively at the level of template vertex sequences and label sequences, proving (a). Because T is rainbow-cycle-free, no nonempty rainbow walk in T can return to its starting vertex u, so $u \notin R_T(u)$, while for each $v \in R_T(u)$ every clone $y \in V_v$ is the terminal vertex of a lifted rainbow walk from $x \in V_u$. Summing over v yields the expressions for $D_G(x)$, $|R_G(x)|$, and $D_G(x) - |R_G(x)| = (Am)_u$. □

3. The 16-Vertex Prototype and Proof of Theorem 1.1

In the 16-vertex, 6-label construction, the capacity matrix $C = (C_{u,v})_{0 \leq u,v \leq 15}$ and the reachability indicator $\mathbf{1}_{\{v \in R_T(u)\}}$ produce the explicit 16×16 integer deficit matrix $A = (A_{u,v})$:

$$A = \begin{pmatrix}
0 & 3 & 2 & 1 & 1 & 1 & 0 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & 0 & -1 \\
0 & 0 & 3 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & 0 & -1 \\
0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & 0 & -1 \\
-1 & -1 & 0 & 0 & 3 & 2 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\
-1 & -1 & -1 & 0 & 0 & 3 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\
-1 & -1 & -1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\
-1 & -1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\
-1 & -1 & -1 & -1 & -1 & -1 & 0 & 0 & 5 & 1 & 1 & 1 & -1 & -1 & -1 & 1 \\
-1 & -1 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & -1 & 1 \\
-1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 0 & 4 & 4 & 1 & 1 & 1 & -1 \\
1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\
1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\
1 & 1 & -1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & 0 & 1 & 1 & 1 \\
1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 0 & 0 & 1 & 2 \\
1 & 1 & 1 & 0 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & 4 \\
1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & 0
\end{pmatrix}.$$

Composition of all $\sum_{\ell=1}^{6} 6!/(6-\ell)! = 1{,}956$ nonempty injective label words over $\{1,\dots,6\}$ on the template T confirms that no diagonal entry is reachable in positive length ($u \notin R_T(u)$ for all $0 \le u \le 15$) and that the off-diagonal reachability relation R_T matches $C - A$.

Evaluating Am on the integer multiplicity vector

$$m = (1096,\ 8,\ 23,\ 69,\ 244,\ 755,\ 1,\ 1059,\ 1,\ 1,\ 1131,\ 1,\ 1,\ 1,\ 1,\ 1062)^{\mathsf{T}}$$

gives the exact deficit vector

$$Am = (1,\ 1,\ 1,\ 1,\ 1,\ 2,\ 1,\ 2,\ 1,\ 213,\ 2,\ 1,\ 1019,\ 128,\ 2182,\ 1)^{\mathsf{T}} \ge \mathbf{1},$$

with $\sum_{u=0}^{15} m_u = 5{,}454$. By Lemma 2.1, the materialized blow-up $G = T[m]$ has 5,454 vertices, $\sum_{u,v} m_u m_v \mathbf{1}_{\{C_{u,v}>0\}} = 11{,}792{,}572$ directed arcs, no rainbow directed cycle, and $D_G(x) - |R_G(x)| \ge 1$ at every vertex $x \in V(G)$. Direct state-reachability traversal on the materialized 5,454-vertex graph independently confirms all 5,454 vertex degrees, reachable sets, and deficits.

Proof of Prototype Optimality (Theorem 1.1(2)). Define the nonnegative integer row-multiplier vector $y \in \mathbb{Z}^{16}_{\ge 0}$ and lower-bound multiplier vector $z \in \mathbb{Z}^{16}_{\ge 0}$ by

$$y = (492,\ 159,\ 45,\ 15,\ 5,\ 0,\ 715,\ 0,\ 694,\ 0,\ 0,\ 738,\ 0,\ 0,\ 0,\ 692)^{\mathsf{T}},$$
$$z = (0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 1207,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 692,\ 0)^{\mathsf{T}}.$$

Direct columnwise summation verifies the exact integer identity

$$A^{\mathsf{T}} y + z = (1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1)^{\mathsf{T}} = \mathbf{1}.$$

For any real vector $m \in \mathbb{R}^{16}$ satisfying $m \ge \mathbf{1}$ and $Am \ge \mathbf{1}$, since $y \ge 0$ and $z \ge 0$, we have

$$\sum_{v=0}^{15} m_v = \mathbf{1}^{\mathsf{T}} m = (A^{\mathsf{T}} y + z)^{\mathsf{T}} m = y^{\mathsf{T}}(Am) + z^{\mathsf{T}} m \ge \sum_{u=0}^{15} y_u + \sum_{v=0}^{15} z_v = 3555 + 1899 = 5454.$$

Since the integer vector m above attains $\sum_{v=0}^{15} m_v = 5454$, both the rational linear program and the integer linear program have exact minimum 5,454. □

4. Scope and Open Boundaries

Theorem 1.1 resolves the multi-label formulation of Conjecture 6.1 in which distinct label graphs G_i may overlap on the same directed arc ($|L(u,v)| \le k$) and reachability is defined by nonempty rainbow walks. Two natural strengthenings remain open:

(i) the *single-label* (edge-disjoint) formulation in which $|L(u,v)| \le 1$ for every ordered pair (u,v), so that $D(v) = d^+_G(v)$;

(ii) determination of the absolute minimum number of vertices required for a multi-label counterexample across all possible templates T.

References

[1] B. D. Sullivan, *A summary of problems and results related to the Caccetta–Häggkvist conjecture*, AIM workshop survey, arXiv:math/0605646, 2006.

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